Data Analytics

Exploratory Data Analysis (EDA)

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- "At a high level, EDA is the practice of using **visual** and **quantitative** methods to understand and summarize a dataset **without making any assumptions about its contents**. It is a crucial step to take **before** diving into machine learning or statistical modeling because it provides the context needed to develop an appropriate model for the **problem at hand** and to correctly interpret its results." Chloe Mawer

Exploratory Data Analysis (EDA)

- “In statistics, exploratory data analysis is an approach of analyzing data sets to summarize their main characteristics, often using **statistical graphics** and other **data visualization methods**.
- Exploratory data analysis has been promoted by John Tukey since 1970 to encourage statisticians to explore the data, and possibly formulate hypotheses that could lead to new data collection and experiments.”

[Wikipedia]()

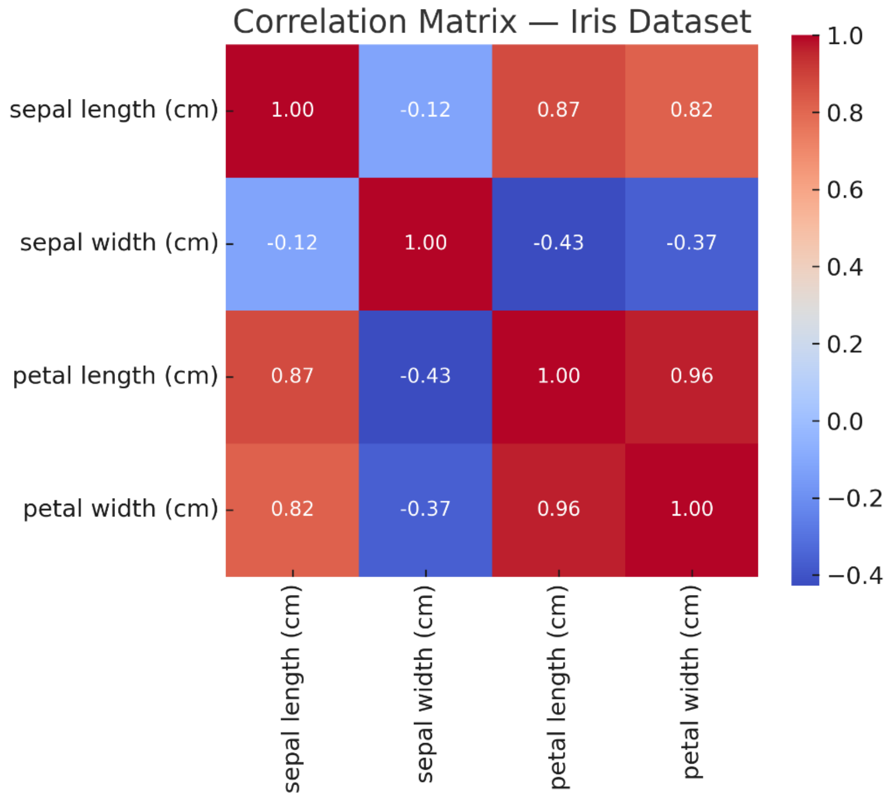
EDA BASICS

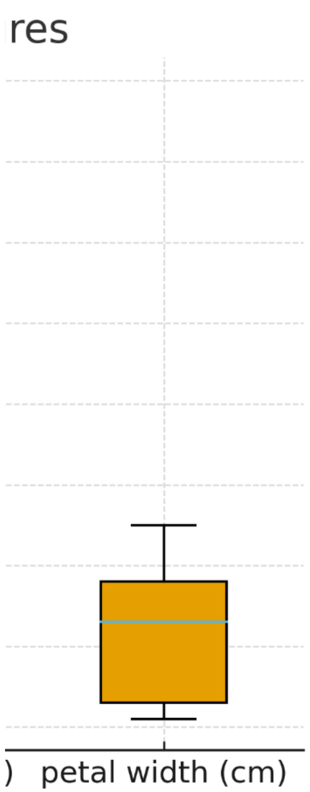
Exploratory Data Analysis (EDA)

- Focus: Exploration before confirmation - look first, model later.
- Supported by tools such as
  - Python: matplotlib, seaborn, pandas
  - R: ggplot2

Exploratory Data Analysis (EDA)

- Typical visual tools: histograms, scatterplots, boxplots, correlation matrices





Five-Number Summary

- Summarise a numerical observation (i.e. a numeric column in a dataset) with five numbers:
  - Minimum
  - First quartile - Q1
  - Median or second quartile – Q2
  - Third quartile – Q3
  - Maximum
- Provides a summary of the distribution of the observation (i.e. the numbers in the column).
- Video example: https://www.youtube.com/watch?v=tpToLyZibKM

Five-Number Summary (contd.)

- There are alternative formulas for calculating Q1 and Q3: https://mathworld.wolfram.com/Quartile.html

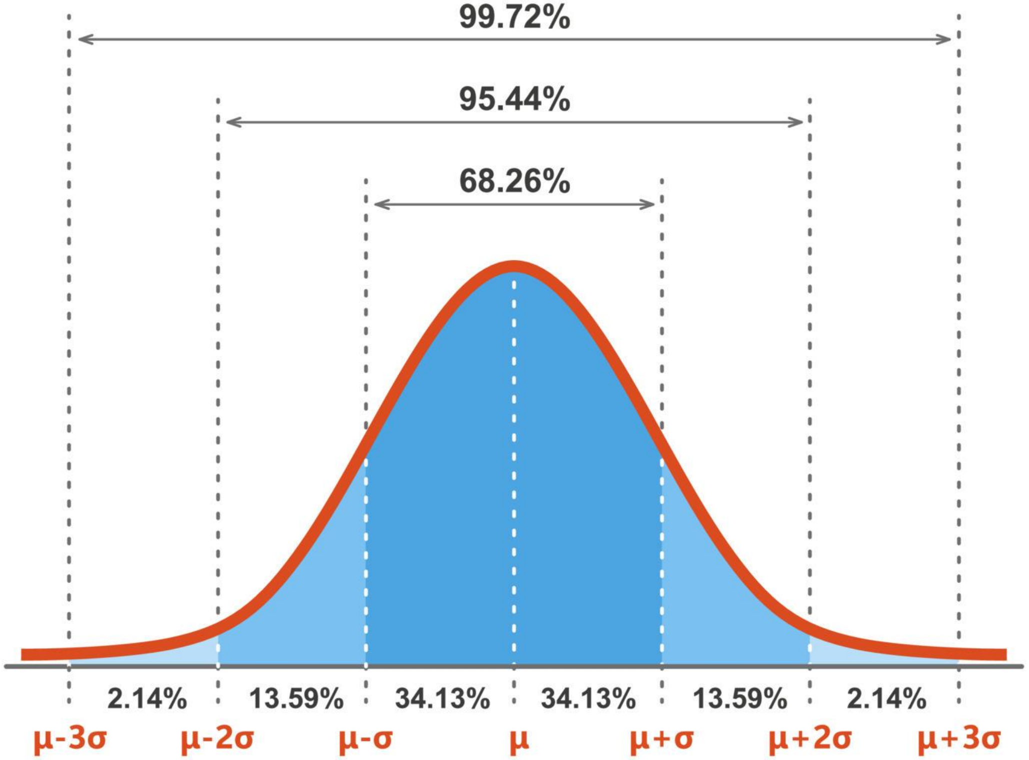
- Interquartile range IQR = Q3 – Q1.

Outliers

- **Variance** of a set of observations - a measure of how spread out the values in a dataset are around the average (mean). It tells us how much the individual numbers differ from the central value, on average.
- **Standard deviation** of a set of observations - the square root of the variance.
- Example: Standard deviation - Wikipedia

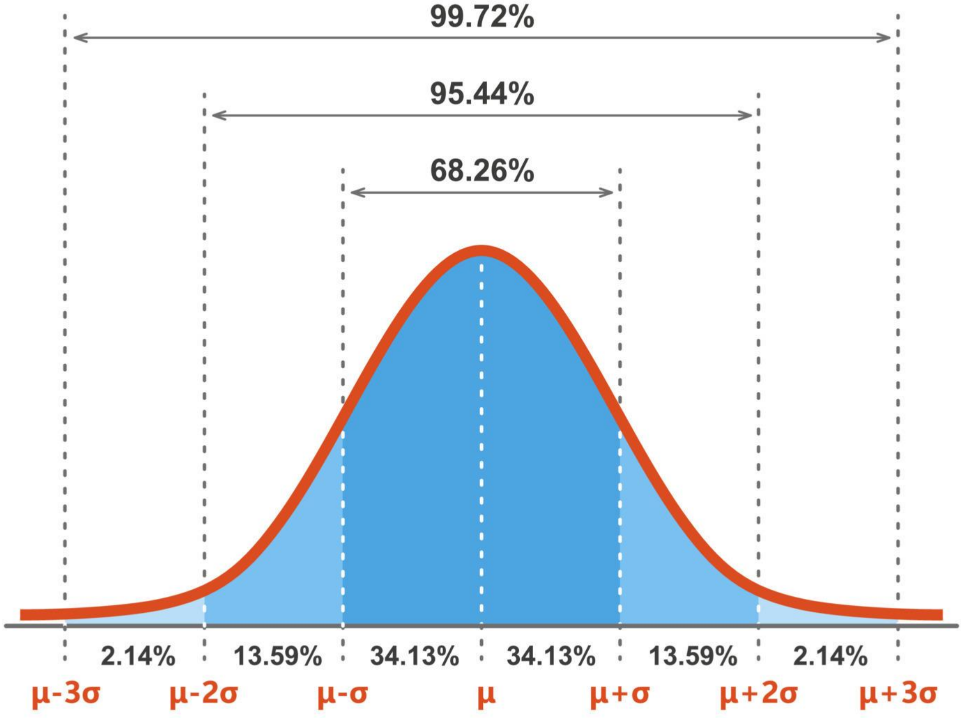
Outliers (contd.)

- Gaussian distribution - Bell-Shaped Curve
- μ - mean, σ - standard deviation



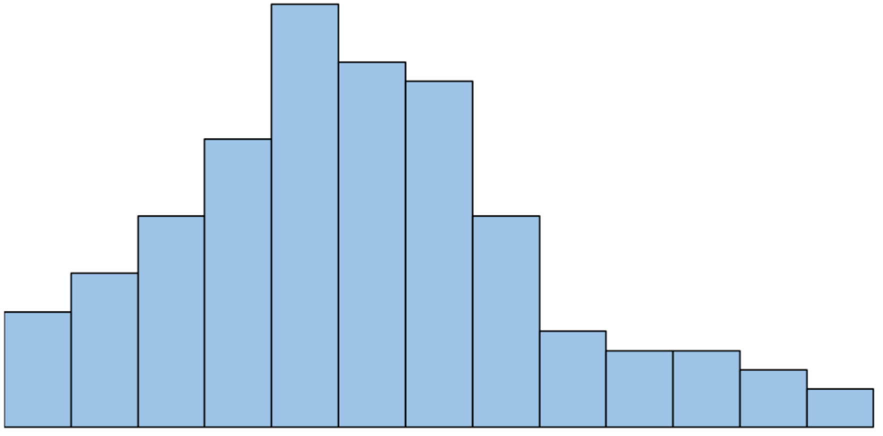
Outliers (contd.)

- Outliers: values below Q1 – 1.5*IQR and above Q3 + 1.5*IQR
- Q1−1.5×IQR and Q3+1.5×IQR roughly capture 99.3% of the data in a Gaussian distribution.

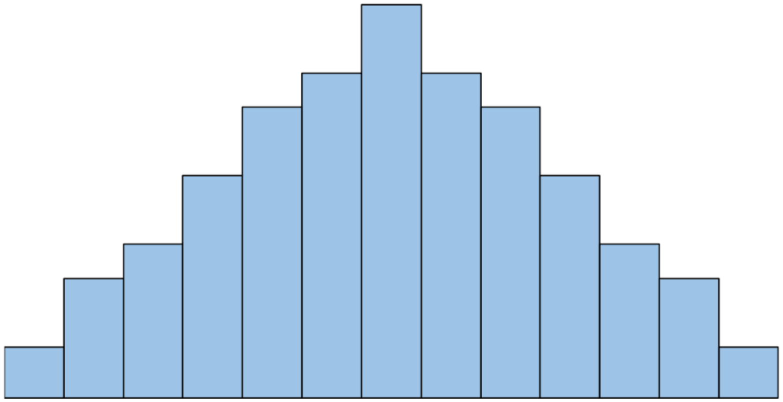


Histograms and Distributions

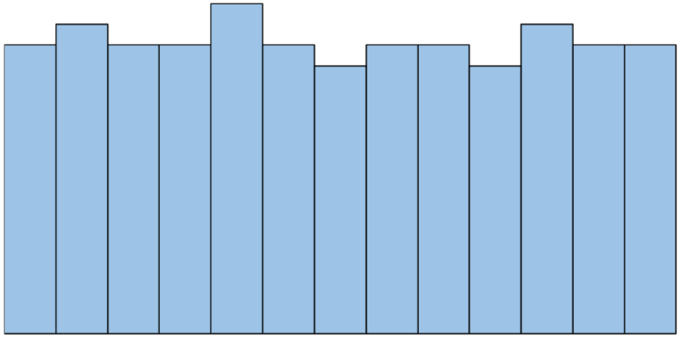
- A histogram is a type of chart that visualises the distribution of values in a set of observations.
- The x-axis displays the values and the y-axis shows the frequency of each value.



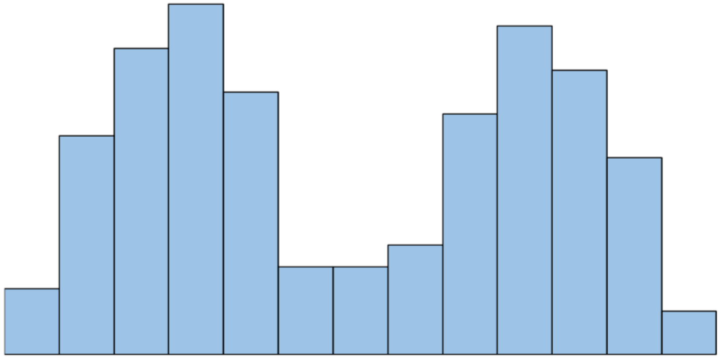
Histograms and Distributions



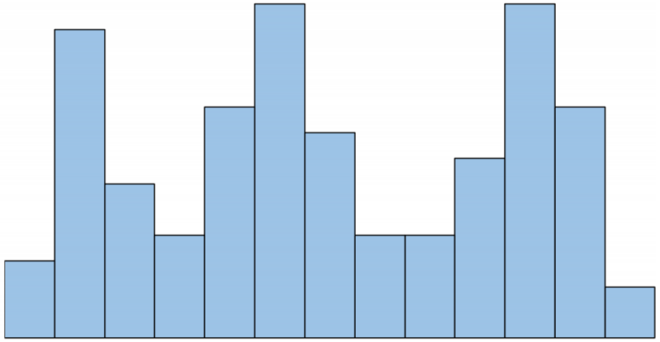
Gaussian/Normal



Uniform



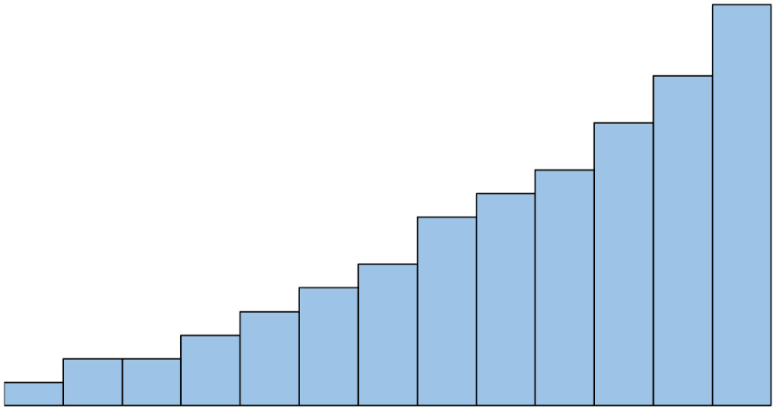
Bimodal



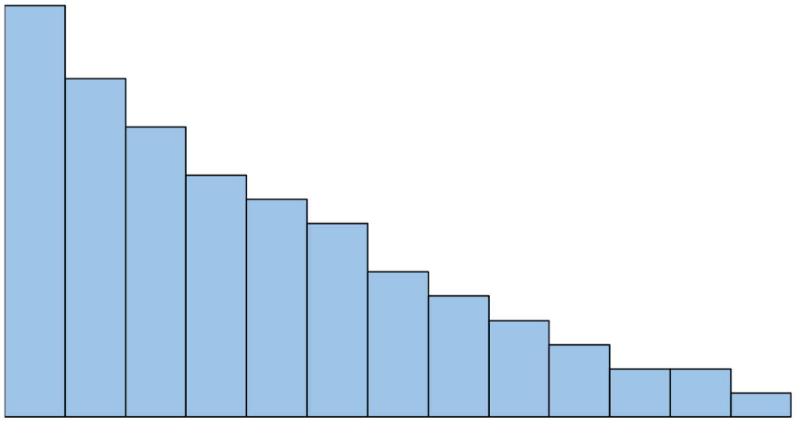
Multimodal

[How to Describe the Shape of Histograms (With Examples)]

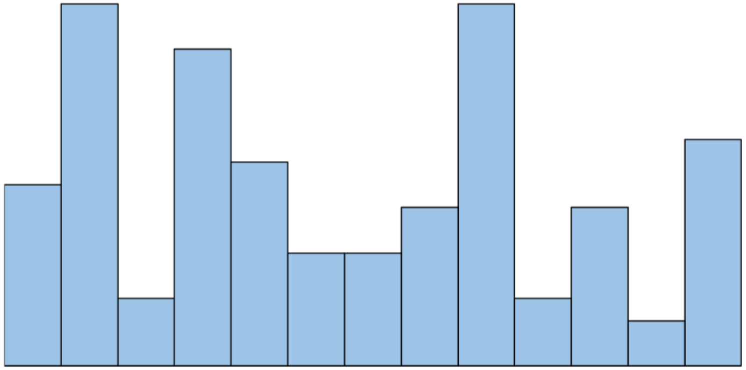
Histograms and Distributions



Left Skewed



Right Skewed



Random

[How to Describe the Shape of Histograms (With Examples)]

EDA BEST PRACTICE

EDA Best Practice

- Explore the data to understand its structure, quality, and potential issues before cleaning and preparing it for machine learning:
- **Check for Sampling Bias** – Ensure the dataset represents the target population adequately.
- **Identify Data Quality Issues** – Detect missing values, outliers, and noise.
- **Recognise Irrelevant Attributes** – Remove non-informative columns (e.g., ID numbers, URLs).
- **Examine Data Distributions** – Assess the statistical properties of each column.

EDA Best Practice (contd.)

- **Analyse Feature Relationships** – Understand dependencies and interactions between variables.
- **Investigate Subgroup Variations** – Identify patterns across different subsets of data.
- **Avoid Over-Interpreting Correlations** – Consider confounding variables before inferring causation.
- **Report All Insights Fairly** – Avoid cherry-picking findings to fit a narrative.
- **Acknowledge Limitations and Assumptions** – Clearly document potential biases and constraints.

EDA Best Practice (contd.)

- Note:
  - In general, EDA comes before data preparation.
  - We may look at portions of the data but we **leave any transformations for the data preparation step that follows**.

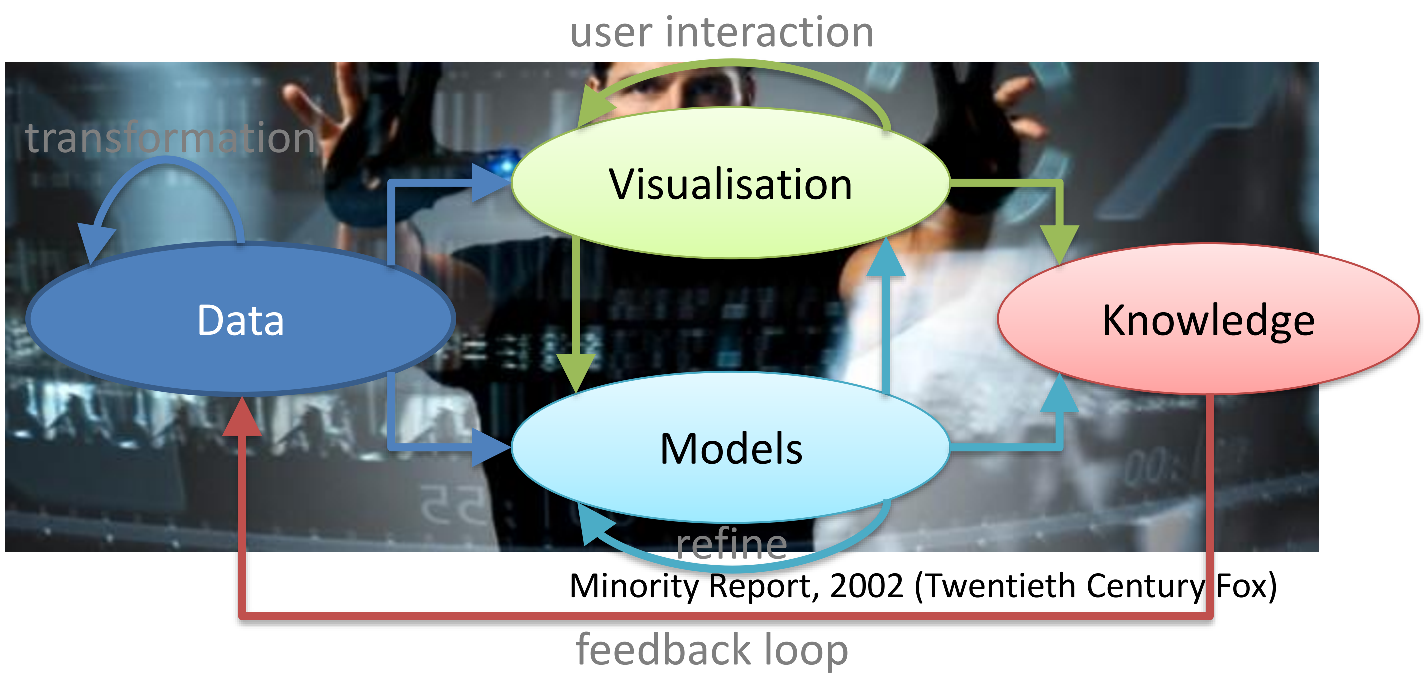
FROM EDA TO VISUAL ANALYTICS

Visual Analytics: EDA for the Age of Big Data

- Originated in the 2000s, merging information visualisation, HCI, and data mining.
- Extends EDA by combining:
  - Interactive exploration: zoom, filter, linked views
  - Automated analysis: ML, clustering, anomaly detection
  - Human reasoning: for insight and decision support.
- Handles large, dynamic, heterogeneous data.
- Tools: Plotly, Tableau, Gephi, D3.js and others.

Visual Analytics

- Visual Analytics is the science of analytical reasoning supported by a highly interactive visual interface (Thomas & Cook, 2005)



Minority Report, 2002 (Twentieth Century Fox)

Visual Analytics Method

- Visual Information Seeking Mantra (Shneiderman, 1996)
  - Overview First
  - Zoom and Filter
  - Details on Demand
- Visual Analytics Mantra (Keim *et al.*, 2010)
  - Analyse First
  - Show the Important
  - Zoom, Filter and Analyse Further
  - Details on Demand

Visual Analytics Challenges

- Intuitively the more data you have, the better…
- Problems when visualising big data:
  - Clutter
  - Performance
  - Information loss
  - Limited cognition